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RADIOACTIVE F-BLOCK ELEMENTS (LANTHANOIDS AND ACTINOIDS): radioactive f-block elements (lanthanoids and actinoids)

Chemistry Quiz: Elements and the Periodic Table

This quiz tests your understanding of the different blocks and groups of elements in the periodic table, based on the provided knowledge base.

Instructions: Read each question carefully and choose the best answer from the options provided.

Question 1: Which block of elements is characterized by the filling of inner d orbitals and is located in the center of the periodic table? A) s-block elements B) p-block elements C) d-block elements D) f-block elements

Answer: C

Question 2: The s-block elements, along with which other block, are collectively known as the representative elements or main group elements? A) d-block elements B) f-block elements C) p-block elements D) Noble gas elements

Answer: C

Question 3: What is a defining characteristic of f-block elements, also known as inner transition elements? A) Their outermost electronic configuration varies from ns^2np^1 to ns^2np^6. B) They are never found pure in nature and their compounds are predominantly ionic. C) The last electron added to each element is filled in an f-orbital. D) They exhibit very low chemical reactivity due to a closed valence shell ns^2np^6 configuration.

Answer: C

Question 4: Which statement is true regarding the noble gas elements? A) They are characterized by the filling of inner d orbitals. B) They have highly negative electron gain enthalpies and readily add electrons. C) They exhibit very low chemical reactivity due to a stable ns^2np^6 configuration. D) They are all radioactive and their chemistry is not fully studied.

Answer: C

Question 5: Based on the provided information, which of the following is true about actinoid elements? A) Their chemistry is simpler than corresponding lanthanoids due to fewer oxidation states. B) They are predominantly ionic compounds, with the exception of those of lithium and beryllium. C) They are all metals, and many are radioactive. D) They are found pure in nature as active metals.

Answer: C

Question 6: The element with atomic number Z = 117, if discovered, would belong to which family/group according to the provided text? A) Alkaline earth metals (Group 2) B) Chalcogens (Group 16) C) Halogens (Group 17) D) Noble gases (Group 18)

Answer: C